SULIT

UTM
UNIVERSITI TEKNOLOGI MALAYSIA

Faculty of Computing

UNIVERSITI TEKNOLOGI MALAYSIA
FINAL EXAMINATION SEMESTER II , 2017 / 2018

SUBJECT CODE	:	**SCSR 1213**
SUBJECT NAME	:	**NETWORK COMMUNICATIONS**
SECTION	:	**ALL**
TIME	:	**9.00 AM – 12. 00 NOON**
DATE/DAY	:	**05 / 06 / 2018 (TUESDAY)**
VENUES	:	**BK 1**

INSTRUCTIONS :

1. Answer ALL questions in the answer booklet.

(Please Write Your Lecture Name And Section In Your Answer Booklet)

Name	
I/C No.	
Year / Course	
Section	
Lecturer Name	

This questions paper consists of NINE (9) printed pages excluding this page.

PART A [10 MARKS]

INSTRUCTION: Please answer all questions in the answer booklet.

1. Computer networks that provide only a connectionless service at the network layer are called _____.

   A) Datagram networks
   B) ATM
   C) Virtual-circuit networks
   D) Frame relay

2. Which address is associated with Network Layer?

   A) Port number
   B) IP address
   C) MAC address
   D) Physical address

3. What type of routing would be best fit to transmit a video to a set of 5 different hosts on a large network?

   A) Unicast
   B) Broadcast
   C) Multicast
   D) Anycast

4. The following processes are involved in the DHCP client-server interaction when a newly arrived host acquires network setting from DHCP server, EXCEPT:

   A) Host broadcasts "DHCP discover" message.
   B) DHCP server responds with "DHCP offer" message.
   C) Host requests IP address using "DHCP request" message.
   D) DHCP server sends address using "DHCP answer" message.

5. Protocol Data Unit (PDU) for Layer 2 is known as _____.

   A) message
   B) segment
   C) frame
   D) datagram

6. How big is the MAC address space?

   A) 2^{16}
   B) 2^{32}
   C) 2^{48}
   D) 2^{128}

7. Which of these is NOT in the ARP table?

   A) IP address
   C) RTT value

B) MAC address

D) TTL value

8. A switch is a device at which layer?

A) Transport

C) Data Link

B) Network

D) Physical

9. Which of these examples is considered multiple hops and infrastructure-based wireless network?

A) Piconet

C) Bluetooth

B) MANET

D) Wifi

10. GSM is a standard developed to describe _____.

A) 1G

C) 3G

B) 2G

D) 4G

PART B [90 MARKS]

INSTRUCTION: Please answer all questions in the answer booklet. Show all your workings.

1. Consider a datagram network using 32-bit host addresses. Suppose a router has four links, numbered 0 through 3, and packets are to be forwarded to the link interfaces as shown in Table 1. **[10 MARKS]**

Table 1: Forwarding table

Destination Address Range	Link Interface
11001000 00010111 00010000 00000000 through 11001000 00010111 00010111 11111111	0
11001000 00010111 00011000 00000000 through 11001000 00010111 00011000 11111111	1
11001000 00010111 00011001 00000000 through 11001000 00010111 00011111 11111111	2
otherwise	3

a) Provide a forwarding table that has four entries, uses longest prefix matching, and forwards packets to the correct link interfaces. [6m]

b) Determines the appropriate link interface for datagrams with the following destination address. Explain briefly how you arrive at that conclusion [4m]

(i) 11001000 00010111 00011000 10101010

(ii) 11101000 11100111 00010110 10100001

2. Consider the topology shown in Figure 1. The three subnets with hosts are Networks A, B, and C. The subnets without hosts are Network D (R1-R3), Network E(R2-E3) and Network E (R1-R2). **[10 MARKS]**

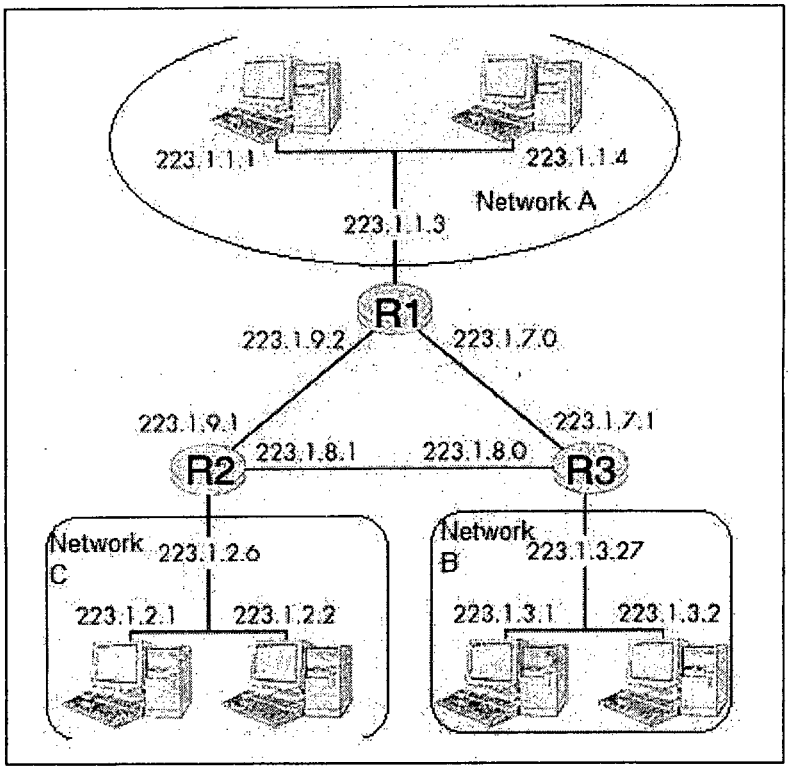


Figure 1: Interconnected network using 3(three) routers

Assign network addresses to each of the six subnets, with the following constraints:

- All addresses must be allocated from 214.97.254.0/23
- Subnet A should have enough addresses to support 250 interfaces
- Subnet B and C should have enough addresses to support 120 interfaces

3. Figure 2 shows fragmented IP datagrams. **[10 MARKS]**

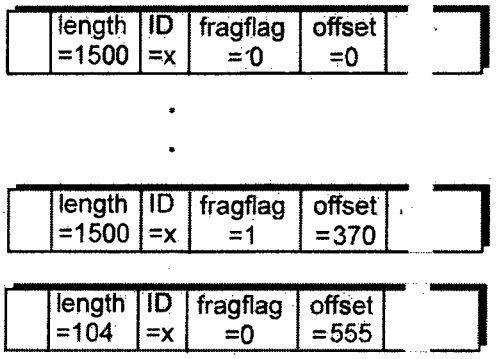


Figure 2: Fragmented IP packet

a) Produce the missing fragmented datagram. [3m]

b) Produce the reassembled IP datagrams at the destination. [5m]

c) What is MTU? What is the size of MTU in the above scenario? [2m]
MTU (max.transfer size) - largest possible link-level frame. 1500 Bytes

4. The following questions relate to error correction. **[10 MARKS]**

a) The polynomial 1001 for CRC code has been selected during the negotiation period of transmitter and receiver. Suppose that a receiver has received frame 11001001010. Should the receiver accept or reject the frame? Justify your answer. [4m]

b) Besides error detection and correction, briefly explain three other services provided by the link layer. [6m]

5. Communication for shared medium are based on multiple access protocols. **[10 MARKS]**

a) What is multiple access protocols? Name the three categories of multiple access protocols and give one example for each category. [4m]

b) How is CSMA/CD different from pure CSMA? [3m]

c) How is CSMA/CD different from CSMA/CA [3m]

6. The following questions refer to LAN communication based on Figure 3, while Table 2 list the IP and MAC addresses of each node. **[10 MARKS]**

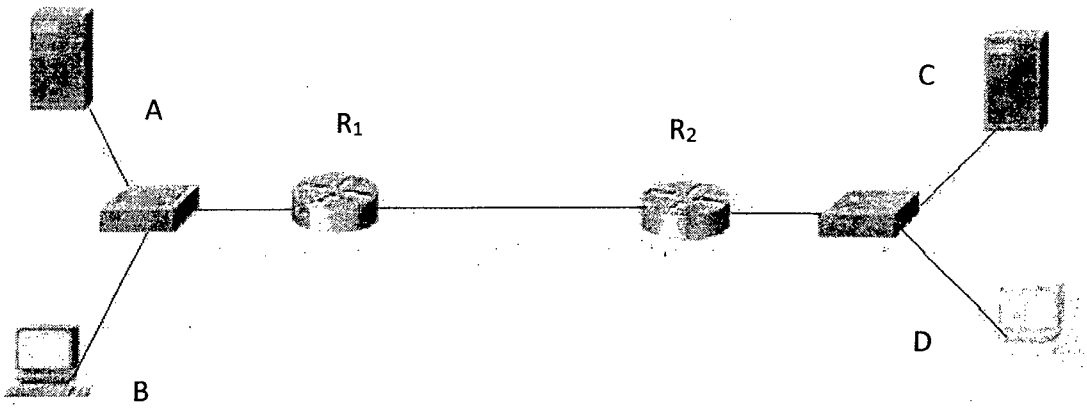


Figure 3: Interconnected LANs

Table 2: IP and MAC address of node

Node	IP	MAC
A	111.111.111.111	AB-12-AB-12-AB-12
B	122.122.122.122	BC-12-BC-22-BC-22
C	133.133.133.133	CD-12-CD-32-CD-32
D	144.144.144.144	DE-12-DE-42-DE-42
R1	155.155.155.155 (left) 166.166.166.166 (right)	AF-12-BA-12-9A-12 AF-12-BB-22-9B-22
R2	177.177.177.177 (left) 188.188.188.188 (right)	B1-12-AA-12-7A-12 B2-12-BB-32-8B-32

a) Suppose Host A wants to send an IP datagram to Host B and it knows B's IP address.

i. Must Host A know B's MAC address to send the datagram to B? [1 m]

ii. If so, how does A get this info? If not, explain why not. [2m]

b) Suppose Host A wants to send an IP datagram to Host C and it knows C's IP address.

i. Must A also know C's MAC address to send the datagram to C? [1 m]

ii. If so, how does A get this info? If not, explain why not. [3m]

c) Suppose that router R_2 has a datagram (that was originally sent by Host A) to send to Host C.

i. What are the MAC addresses on the frame that is sent from router R_2 to Host C? [2m]

ii. What are the IP addresses in the IP datagram encapsulated within this frame? [2m]

7. There are 6 nodes labelled A through F connected into an Ethernet switch. Reproduce Table 3 with answers into your answer booklet. Show the state of the switch table before and after each of the events. For each event, identify the link(s) on which the transmitted frame will be forwarded, and briefly justify your answer. **[10 MARKS]**

Table 3: Switch state and action

	Action/ Event	**Switch Table State**	**Link(s) packet is forwarded to**	**Explanation**
a)	B sends a frame to E			
b)	E replies with a frame to B			
c)	A sends a frame to B			
d)	B replies with a frame to A			

8. Figure 5 shows 3G cellular data network architecture. Choose required components that will enable a mobile client to connect to Internet. Provide description for each of the chosen components. **[10 MARKS]**

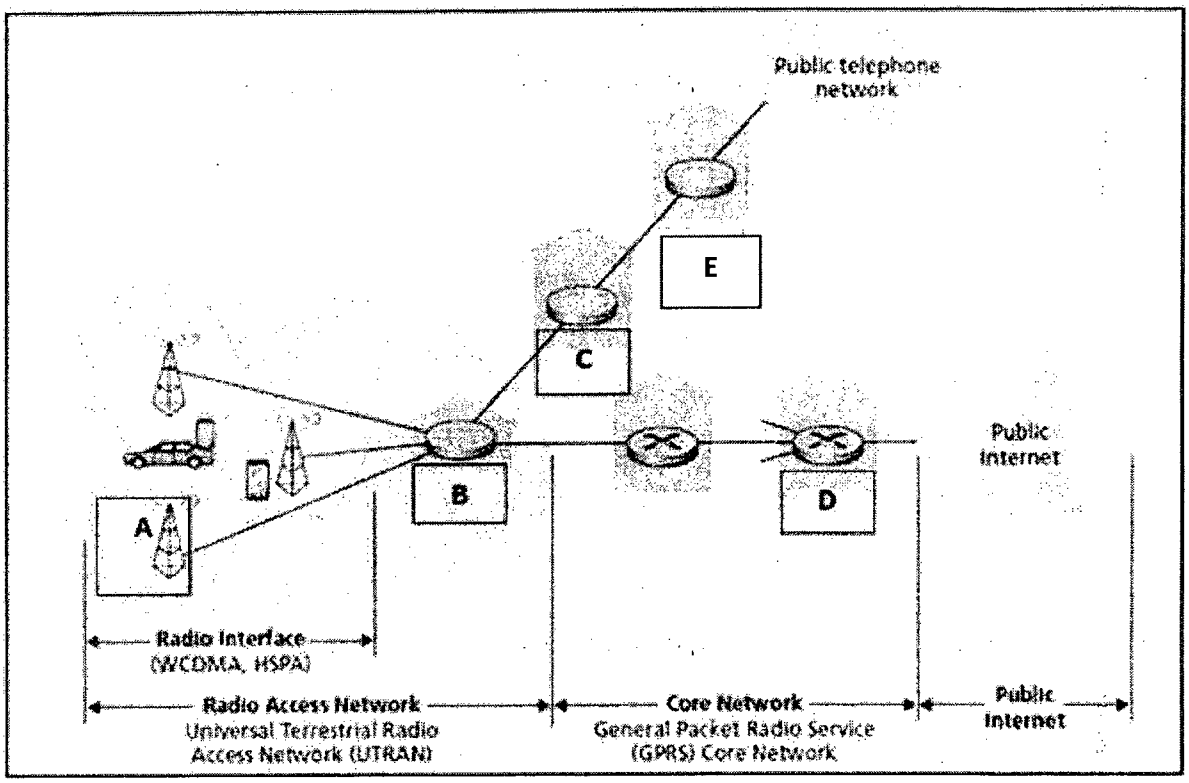


Figure 5: 3G system architecture

9. Referring to Figure 6, node A moves from home network to visited network. **[10 MARKS]**

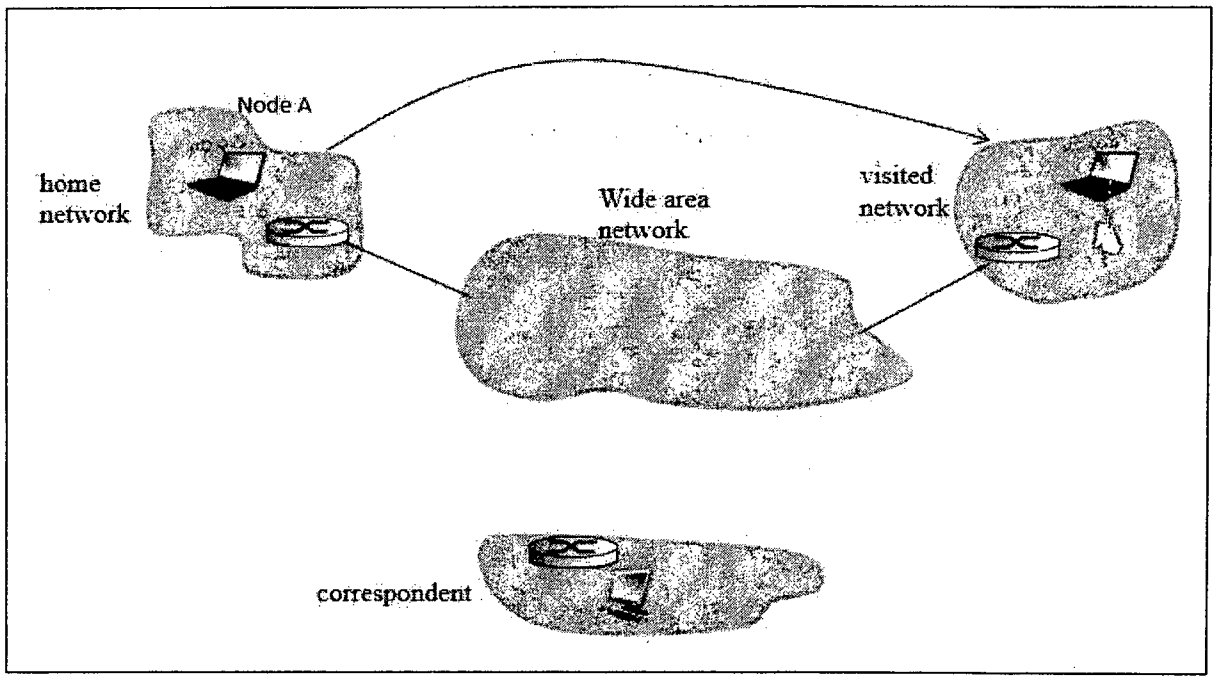


Figure 6: Mobile IP

a) Briefly describe permanent and care-of addresses. [4m]

b) Draw arrows (label with sequence number) to illustrate the interaction of node A and corresponding node, when corresponding node is to transfer datagram to Node A (using either indirect or direct routing). [6m]